

AI AMC 4.0: Session 7

RAG Evaluation: Search Evaluator & Answer Evaluator

Building reliable RAG pipelines requires rigorous evaluation at every stage — from retrieval quality to answer accuracy. This session equips you with the full toolkit.



Why RAG Needs Evaluation

Without evaluation, a RAG pipeline is essentially a black box: a user query goes in, documents are retrieved, an answer is generated — and you have no idea if any of it was correct.

? Did we retrieve the right documents?

Irrelevant sources silently poison the generation step before it even begins.

🔗 Is the answer grounded in sources?

An answer that ignores retrieved context defeats the entire purpose of RAG.

🧠 Did the LLM hallucinate?

LLMs confidently fabricate facts — without a check, hallucinations reach users undetected.

✅ Is the answer actually helpful?

Vague, evasive responses erode user trust even when technically "not wrong."

- ✅ RAG **with evaluation** adds two critical checkpoints: evaluate retrieval quality **before** generation, and evaluate answer quality **after** generation — enabling auto-retry, hallucination detection, and quality monitoring.

Two Pillars of RAG Evaluation

Every robust RAG evaluation framework rests on two complementary pillars — one guarding the retrieval stage, the other guarding the generation stage.



Search Evaluation catches bad retrieval before it can corrupt generation. Answer Evaluation catches hallucinations and vagueness after the LLM has responded. Together, they form a full-coverage quality gate across the entire pipeline.

Search Evaluation — The Core Question

SEARCH METRICS

"Did we retrieve the right documents?" — Consider this example query: *"What is Apple's Q3 2024 revenue?"* The retriever returns 5 documents. Not all are equally useful.

#	Retrieved Source	Relevant?	Why
1	Apple Q3 2024 Earnings Report	✔ Yes	Directly answers the query
2	Apple Stock Price Today	✗ No	Current price ≠ quarterly revenue
3	Apple Q2 2024 Earnings Report	◐ Partial	Wrong quarter
4	Microsoft Q3 2024 Report	✗ No	Wrong company entirely
5	Apple Revenue History 2020–24	✔ Yes	Contains Q3 2024 data

- **Are relevant docs ranked at top?**
→ Measured by **MRR** and **NDCG**
- **How many relevant docs did we find?**
→ Measured by **Precision** and **Recall**
- **Overall retrieval quality?**
→ Measured by **MAP** and **F1**

Mean Reciprocal Rank (MRR)

SEARCH METRIC

Core Question

How quickly do we find the **first relevant document**? MRR rewards systems that surface useful results immediately — before users have to scroll.

📄 **Formula:** $MRR = 1 / \text{rank_of_first_relevant_document}$

Worked Example

Result 1: ❌ Not relevant

Result 2: ✅ **First relevant — at position 2**

Result 3: ❌ Not relevant

Result 4: ✅ Relevant

$MRR = 1/2 = 0.50$

Interpretation Guide

1.0 — Perfect

First result is always relevant

0.5 — Acceptable

Relevant doc typically at position 2

0.1 — Poor

Users scroll through 10+ irrelevant results

✅ 🎯 **Target: $MRR > 0.8$** — Users expect the first result to be useful. Anything less degrades chatbot experience.

Precision & Recall

SEARCH METRICS

Precision and Recall are complementary lenses on retrieval quality — one measures accuracy, the other measures coverage.

Precision — "Are retrieved docs useful?"

📄 **Formula:** Precision = Relevant Retrieved / Total Retrieved

We retrieved 5 docs: 3 are relevant, 2 are not.

Precision = $3/5 = 0.60$

60% of what we returned is actually useful.

✅ 🎯 Good Precision: > 0.8

Recall — "Did we find all relevant docs?"

📄 **Formula:** Recall = Relevant Retrieved / Total Relevant in DB

Database has 10 relevant Apple Q3 2024 documents. We found only 3.

Recall = $3/10 = 0.30$

We missed 70% of all relevant documents!

✅ 🎯 Good Recall: > 0.7

⚠️ ⚠️ High Precision with Low Recall means your results are accurate but incomplete — you're missing crucial information. Both metrics must be strong for a robust retrieval system.

F1 Score — Balancing Precision & Recall

SEARCH METRIC

The Harmonic Mean Formula

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F1 uses the **harmonic mean** — not arithmetic — because arithmetic mean lets high precision *mask* catastrophically low recall. The harmonic mean punishes imbalance severely.

✔️ 🎯 Good F1: > 0.7

Why F1 Exposes Hidden Problems

System	Precision	Recall	F1	Verdict
A	0.9	0.1	0.18	❌ Poor Balance
B	0.6	0.6	0.60	✅ Balanced
C	0.8	0.8	0.80	✅ Strong

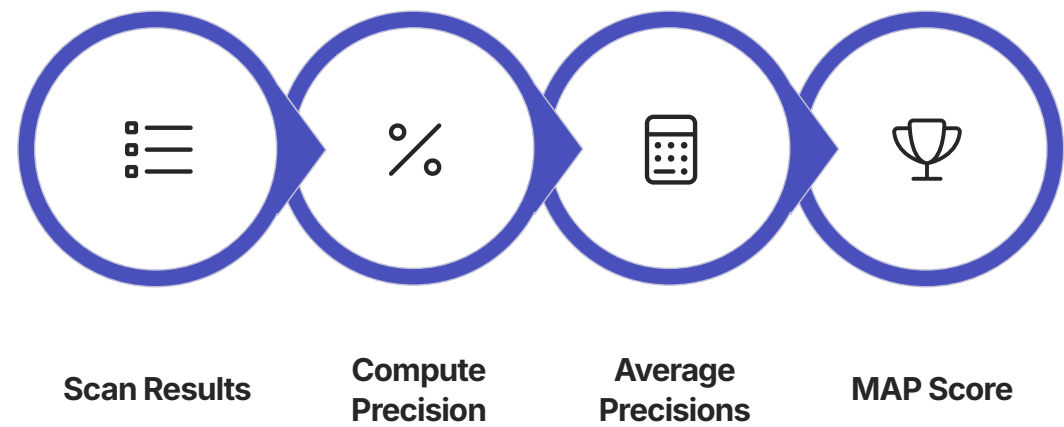
❌ System A's arithmetic mean = 0.5 — which looks passable. F1 = 0.18 reveals the truth: it almost never finds relevant documents!

Mean Average Precision (MAP)

SEARCH METRIC

MAP asks: **Across ALL relevant documents, how good is our overall ranking?** Unlike MRR (which stops at the first hit), MAP evaluates every relevant document's position.

Step-by-Step Algorithm



Worked Example: Results = [X, ✓, X, ✓, ✓, X, X, ✓, X, X]

Position	Relevant?	Precision at Point	Value
2	✓	1 relevant / 2 retrieved	0.50
4	✓	2 relevant / 4 retrieved	0.50
5	✓	3 relevant / 5 retrieved	0.60
8	✓	4 relevant / 8 retrieved	0.50

MAP = (0.50 + 0.50 + 0.60 + 0.50) / 4 = 0.525 — Systems that front-load relevant docs score significantly higher. 🎯 Good value: > 0.7

Normalized Discounted Cumulative Gain (NDCG)

SEARCH METRIC

NDCG answers: **How good is our ranking, given that position matters enormously?** A relevant document at rank 1 is worth far more than the same document buried at rank 10.

Position Discount ($\log_2$)

 100%

Position 1

Full credit

 63%

Position 2

Discounted

 50%

Position 3

Half credit

 39%

Position 5

Heavily discounted

Worked Example

Actual ranking: [0, 1, 0, 1, 1] vs. Ideal: [1, 1, 1, 0, 0]

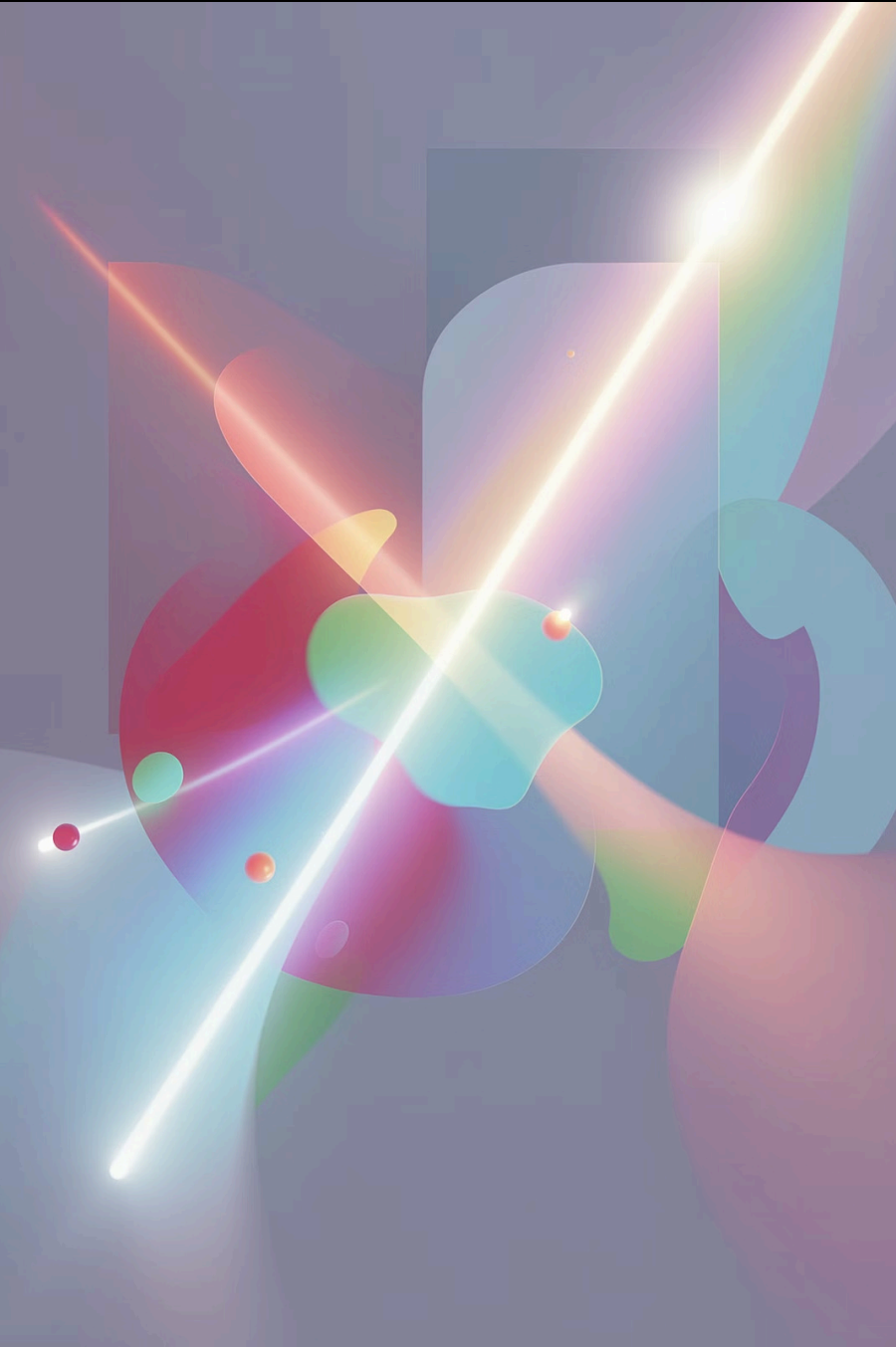
$$DCG_{actual} = \frac{1}{\log_2 3} + \frac{1}{\log_2 5} + \frac{1}{\log_2 6} = 1.45$$

$$DCG_{ideal} = \frac{1}{\log_2 2} + \frac{1}{\log_2 3} + \frac{1}{\log_2 4} = 2.13$$

$$NDCG = \frac{1.45}{2.13} = 0.68$$



🎯 Good NDCG: **> 0.8** — NDCG normalizes against the ideal ranking, making it comparable across queries.



Answer Evaluation — The Core Question

ANSWER METRICS

"Is the generated answer actually good?" Retrieval quality is only half the battle. Even with perfect sources, the LLM can still produce wrong, vague, or fabricated answers.



Source (Ground Truth)

*"Apple reported Q3 2024 revenue of **\$85.5 Billion**, up **5%** year-over-year."*



Generated Answer

"Apple's Q3 2024 revenue was 85.5 billion, representing a 5% increase compared to the same quarter last year."

Word Overlap

ROUGE, Fuzzy Matching

Grounding

LLM Source Check

Hallucination

LLM Precision (TP/FP)

Helpfulness

LLM Relevancy Check

ROUGE — Recall-Oriented Understudy for Gisting Evaluation

ANSWER METRIC

Originally designed to evaluate text summarization, ROUGE measures **how much the generated text overlaps with a reference text** at the word level.

ROUGE-1: Unigram (Word) Overlap

Reference: "Apple reported strong quarterly earnings"

Generated: "Apple announced strong quarterly results"

Reference Unigrams	Generated Unigrams
Apple, reported, strong, quarterly, earnings	Apple, announced, strong, quarterly, results

✔ Matching = {Apple, strong, quarterly} = **3 words**



ROUGE-2 and ROUGE-L

ANSWER METRICS

ROUGE-2: Bigram (Phrase) Overlap

Counts matching **word pairs**, capturing phrase-level similarity rather than isolated word matches.

Reference Bigrams	Generated Bigrams
Apple-reported	Apple-announced
reported-strong	announced-strong
strong-quarterly	strong-quarterly ✓
quarterly-earnings	quarterly-results

Matching = {strong-quarterly} = **1 pair**

$$ROUGE-2\ F1 = \frac{2 \times (1/4 \times 1/4)}{1/4 + 1/4} = 0.25$$

✓ 🎯 Good ROUGE-2: > 0.3

ROUGE-L: Longest Common Subsequence

Finds the **longest sequence of words appearing in the same order** across both texts — not necessarily consecutive. This captures sentence-level structure, rewarding answers that preserve the flow of the reference even when exact phrases differ.

- More flexible than bigram matching
- Handles paraphrasing better than ROUGE-2
- Penalizes reordered information
- Best proxy for overall answer coherence

✓ 🎯 Good ROUGE-L: > 0.4

Fuzzy String Matching

ANSWER METRIC

Fuzzy matching handles the real world: typos, abbreviations, and extra context that word-overlap metrics misinterpret. It uses **Levenshtein Distance** — the minimum number of edits (insert, delete, replace) to transform one string into another.

Two Key Flavors

fuzz.ratio

Overall similarity score (0–100) based on the full string comparison. Sensitive to extra words.

fuzz.partial_ratio

Finds the best matching **substring**. Crucial when the answer contains extra context around the key fact.

Levenshtein example:

"Apple Inc" → "Apple Inc." = distance 1

Similarity = $1 - (1/10) = 90\%$

Why Partial Ratio Wins

Reference: "Revenue was \$85.8 billion"

Generated: "In Q3 2024, Apple's revenue was \$85.8 billion, up 5% YoY."

~40

Full Ratio

Extra words heavily penalize the score

~95

Partial Ratio

Key phrase "\$85.8 billion" matches perfectly



Good Fuzzy Score: > 70

Semantic Similarity (Embeddings)

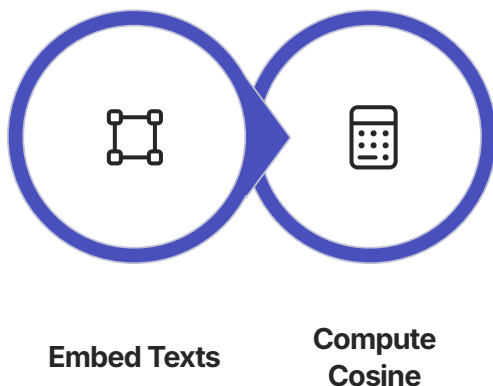
ANSWER METRIC

Semantic similarity asks: **Do these texts mean the same thing, even if they share zero words?** This is the critical failure mode of all word-overlap metrics.

The Word-Overlap Failure Case

✗ Text A: "The stock price has increased"
Text B: "The shares went up"
Word overlap = 0% — ROUGE gives a score of 0!
But semantically, these are *identical*.

How Embeddings Fix This



Embedding Similarity in Action

Text A → [0.2, 0.8, 0.1, ...]

Text B → [0.3, 0.7, 0.2, ...]

0%

Word Overlap

ROUGE completely fails

92%

Cosine Similarity

Embeddings capture true meaning

✓ This is why embeddings are the backbone of modern RAG — they capture **meaning**, not just surface-level word patterns.

Why LLM-as-a-Judge?

The Blind Spots of Traditional Metrics

ROUGE and Fuzzy Matching are fast and deterministic — but they have a critical flaw: they measure **surface form, not meaning or factual accuracy**.

❌ Case 1 — Low ROUGE, But Actually Correct

Reference	"Apple's revenue was \$85.8 billion"
Generated	"Apple reported earnings of \$85.8B"
ROUGE-1	Low score — different words!
Human Verdict	✅ These mean the same thing

💣 Case 2 — High ROUGE, But Factually WRONG

Reference	"Apple's revenue was \$85.8 billion"
Generated	"Apple's revenue was \$85.2 billion"
ROUGE-1	Very high — most words match!
Human Verdict	❌ Factually wrong! \$85.8 ≠ \$85.2

👉 **✅ Solution:** Use an LLM as a judge — it understands semantic meaning, catches numerical errors, and evaluates answers the way a human expert would.

LLM Grounding Check

LLM-AS-A-JUDGE

Question: Is each retrieved source actually useful for supporting the final answer? The grounding check forces the LLM judge to rate every source independently — revealing how much of your retrieval is genuinely contributing.

Example — Answer: "Apple's Q3 revenue was \$85.8B, up 5% YoY."

Source 1

"Apple Q3 2024 earnings: \$85.8B revenue, +5% growth"

→ **Useful: 1**  Directly supports the answer with exact figures.

Source 2

"Apple announces new iPhone model"

→ **Useful: 0**  Not relevant to revenue question at all.

Source 3

"Tech sector sees mixed Q3 results"

→ **Useful: 0**  Too generic — provides no specific support.

33%

Grounding Score

Only 1 of 3 retrieved sources actually helped generate the answer



 Good Grounding: **> 80%** — Low grounding means your retriever is returning noise. Fix the retriever, not the generator.

LLM Precision Check — Hallucination Detection

LLM-AS-A-JUDGE

Question: Did the LLM add information **NOT** present in the sources? Every claim in the generated answer is classified as TP, FP, or FN.

TP — True Positive Claim in answer IS supported by sources. Correct and grounded.	FP — False Positive 🚨 Claim in answer is NOT in sources . This is hallucination . The LLM invented it.	FN — False Negative Important information in sources is missing from the answer . Incomplete response.
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Worked Example

Source: "Apple Q3 revenue: \$85.8B"

Generated: "Apple reported Q3 revenue of \$85.8B. Mac sales declined by 3%."

Claim	Classification	Verdict
"Q3 revenue of \$85.8B"	TP	✅ Supported by source
"Mac sales declined by 3%"	FP 🚨	❌ HALLUCINATION — not in any source!

Result: TP=1, FP=1, FN=0 → **Precision = 1/2 = 50%**

❌ 🚨 ALERT: Answer contains hallucination! 🎯 Target: TP% > 90%, FP% < 10%

LLM Relevancy / Non-Committal Check

LLM-AS-A-JUDGE

Question: Is the answer actually helpful, or is it vague and evasive? An LLM that hedges constantly — even when the answer is clearly available in the sources — is failing the user.

Non-Committal Patterns to Detect



"I don't know"



"It depends"



"I am not sure"



"I believe it is..."



"You might want to check..."

Query: "What was Apple's Q3 revenue?"



✓ Answer A — Direct

"Apple's Q3 2024 revenue was \$85.8 billion."

LLM Judge: noncommittal = 0

Relevancy = 1.0



✗ Answer B — Evasive

"Apple's revenue varies by quarter. You might want to check their investor relations page."

LLM Judge: noncommittal = 1

Relevancy = 0.2



🎯 Target: Relevancy = 1 — When sources contain the answer, the LLM must commit to it directly and specifically.

Metrics Quick Reference — Cheat Sheet

A complete reference of all RAG evaluation metrics covered in this session, with ranges and target thresholds.

Search Evaluation Metrics

Metric	Range	Target	Measures
MRR	0–1	>0.8	Speed to first relevant doc
MAP	0–1	>0.7	Overall ranking quality
Precision	0–1	>0.8	% of retrieved docs that are relevant
Recall	0–1	>0.7	% of relevant docs found
F1	0–1	>0.7	Balance of precision & recall
NDCG	0–1	>0.8	Ranking quality with position weighting

Traditional Answer Metrics

Metric	Range	Target	Measures
ROUGE-1	0–1	>0.5	Unigram word overlap
ROUGE-2	0–1	>0.3	Bigram phrase overlap
ROUGE-L	0–1	>0.4	Longest common subsequence
Fuzzy	0–100	>70	String similarity with typo tolerance

LLM-as-a-Judge Metrics

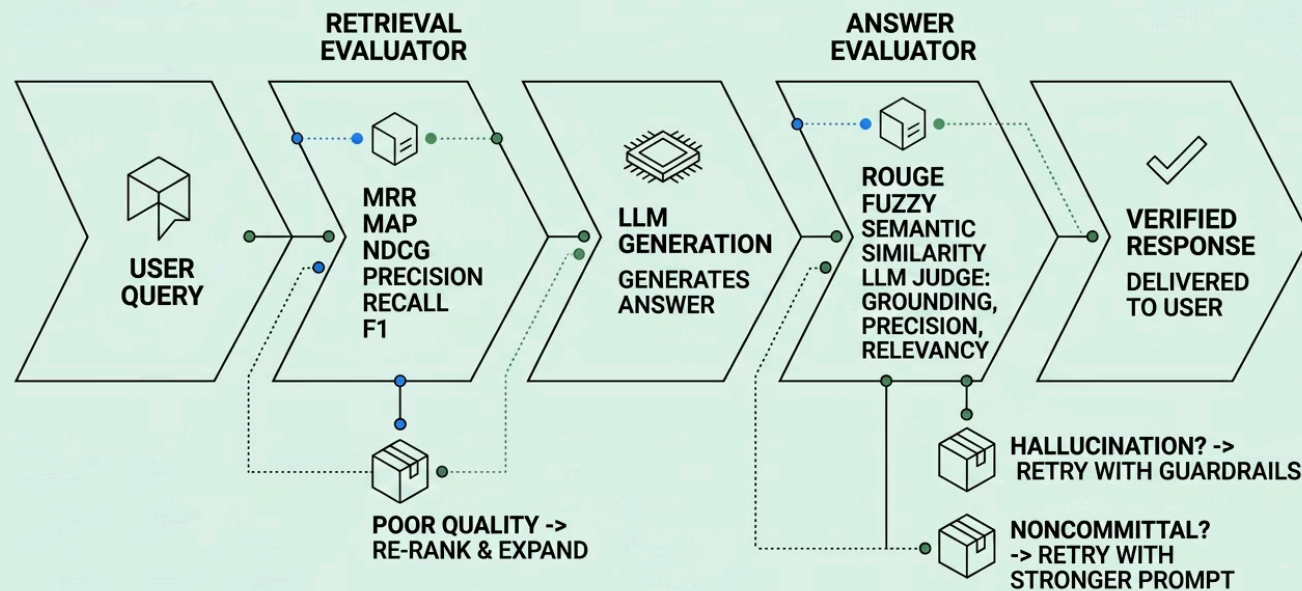
Metric	Range	Target	Measures
Grounding %	0–100%	>80%	Sources that support the answer
TP %	0–100%	>90%	Claims supported by sources
FP % 	0–100%	<10%	Hallucinated claims
Relevancy	0 or 1	=1	Answer is direct, not evasive

 **FP% is the most critical metric** — even a single hallucinated claim can destroy user trust in a production RAG system.

 **✓ No single metric tells the whole story.** Run all four LLM-judge checks together for complete coverage.

Putting It All Together

The Complete RAG Evaluation Pipeline



Catch Bad Retrieval Early

Stop bad sources before they reach the LLM. Fix the retriever, save the generation.

Detect Hallucination Late

LLM judge catches fabricated claims after generation, before they reach users.

Enable Auto-Retry

Low scores trigger automated re-ranking, query expansion, or prompt strengthening.

Build Quality Dashboards

Continuous metric tracking enables systematic improvement over time.

"Learn to build agents before agents learn to replace you."